



Unit 4 · Outcome 2 · Energy Sources

The mine stops. Now someone has to heal the hole.

KK08

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Why we can't just walk away

01 Why we can't just walk away A decommissioned energy site isn't neutral ground — it is actively dangerous and degraded. Rehabilitation exists to remove that hazard and hand the land back with a future. Picture the fictional Yarrabin open-cut, the brown-coal pit that fed the Yarrabin Energy Park for sixty years. When the last shovel-load is burned, it leaves an 18-kilometre-perimeter hole up to 135 metres deep, ringed by near-vertical walls of soft clay and coal. Left alone, that site slumps, erodes, catches fire, and leaches acid and metals into ground

Two toolkits: mechanical & biological

02 Two toolkits: mechanical & biological VCAA splits rehabilitation into exactly two process families. Getting a clean definition of each — and never blurring them — is the core skill this dot point tests. Mechanical · physical & engineering Machines move the earth Non-living, engineering processes that reshape and stabilise the landform and lock away hazards. Bulldozers, excavators, dump trucks and graders do the work. Examples: battering down (flattening) steep highwalls to a stable slope; backfilling the void with overburden/waste rock; capping expose

Inside a rehabilitated landform

03 Inside a rehabilitated landform A rehabilitated site is built up in engineered layers. Peel them back to see where the mechanical work ends and the biological work begins. LIVE 3D drag to orbit · click a layer 3D layer cross-section (WebGL unavailable — use the layer buttons below). Full profile — 5 engineered layers Full profile Vegetation Topsoil Subsoil cap Clay seal Backfill The profile, bottom to top: waste-rock backfill rebuilds the bulk landform (mechanical) → a low-permeability clay seal caps the coal so it can't burn or leach (mechanical) → s

Sort the processes

◇ ACTIVE RECALL

04 Sort the processes Drag each process into the correct family. These are exactly the kind of items examiners list and ask you to classify — get them automatic. Mechanical Biological s

Why the order is non-negotiable

05 Why the order is non-negotiable A favourite exam discriminator: explaining the sequence. Mechanical work almost always comes first — and there's a clear cause-and-effect reason. You cannot grow a forest on a 70° wall of slumping clay. The mechanical stage must come first to create a landform that won't move: slopes are battered back to a stable angle, the void is backfilled, drainage is shaped, and the hazardous coal is sealed under clay. Only once the ground is safe and stable does the biological stage make sense — because revegetation needs a surface

Bringing the land back to life

06 Bringing the land back to life The biological stage is slow, living and measurable. Watch a stabilised surface move through ecological succession — and see how plants can actually clean the ground. LIVE 3D drag to orbit 3D revegetation / succession scene (WebGL unavailable — the year slider still drives the readout). Year 0 · bare respread topsoil Years since revegetation 0 yr Succession Phytoremediation Succession. Pioneer grasses and nitrogen-fixers colonise first, building soil organic matter; shrubs follow; then canopy trees. Returning soil microb

Rehabilitate is not restore

07 Rehabilitate is not restore This distinction is quietly worth marks. Examiners punish students who treat the two as synonyms. Rehabilitation Returning the land to a safe, stable and sustainable condition fit for a viable land use — which may be different from the original (e.g. a former pit becomes farmland, a recreation lake, or a solar farm). The bar is function and safety, not historical accuracy. Restoration The more ambitious goal of returning the site to its original, pre-disturbance ecosystem — same species, same structure, same function. Far h

The Latrobe Valley — Australia's real test case

08 The Latrobe Valley — Australia's real test case Yarrabin is fictional; the science and the stakes are not. Victoria is right now attempting the largest mine rehabilitation in its history, on our doorstep in Gippsland. Three giant brown-coal mines — Hazelwood (closed 2017), Yallourn (closing 2028) and Loy Yang (around 2035) — have powered Victoria since the 1920s. Hazelwood alone is an 18 km-perimeter void up to 135 m deep, with rehabilitation estimated at roughly \$743 million . The Hazelwood mine is bigger than the Melbourne CBD; cleaning it up is the

How it ties to the sustainability principles

09 How it ties to the sustainability principles KK08 doesn't live in isolation — examiners love linking rehabilitation back to the principles you met earlier in the course. Intergenerational equity Rehabilitating now means we don't dump a dangerous, useless landscape on future generations. An un-rehabilitated pit is a debt passed forward. Conservation of ecological integrity Revegetation, bioremediation and reconnecting habitat rebuild functioning ecosystems and biodiversity on degraded land. User-pays principle Rehabilitation bonds and trailing-liabilit

Command-word decoder

◆ COMMAND WORDS

10 Command-word decoder The verb in the question tells you what shape your answer must take. Tap each to see what the examiner actually wants for this topic.

Six exam traps

⚠ EXAM TRAPS

11 Six exam traps The predictable ways students drop marks on this dot point. Open each before you sit a SAC.

P·E·L writing trainer

P·E·L WRITING

12 P·E·L writing trainer P oint · E vidence/example · L ink. Write a paragraph and watch live which key ideas you've actually hit. Aim to light up every chip. POINT State the process clearly EVIDENCE Name a real example LINK Tie back to the goal Start typing to light up the key ideas examiners look for.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Worked answer: weak vs strong

13 Worked answer: weak vs strong Same question, two responses. Read why the examiner's pen moves — then steal the strong version's structure. Question (4 marks): Describe one mechanical and one biological process used to rehabilitate a former open-cut coal mine, and explain how each contributes to a safe, stable and sustainable landform. **X** Weak response — 1/4 They fix up the mine by cleaning it and planting some trees so it looks nice again and animals can come back to live there like before . "Cleaning it" isn't a named mechanical process — no reshapi



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice exam · 22 marks

★ PRACTICE & SELF-MARK

14 Practice exam · 22 marks Five VCAA-style questions. Attempt each in your logbook first, then open the model answer and self-mark honestly with the stars. Your score builds in the dock.

One-screen cheat sheet

✦ RECAP / CHEAT SHEET

15 One-screen cheat sheet KK08 — Rehabilitating energy sites, at a glance Mechanical (machines, non-living) Reshape / batter steep walls to stable slope Backfill void with overburden/waste rock Cap / seal coal with clay (stop fire & leaching) Deep-rip to relieve compaction Contour drainage; build erosion banks Plug / cap oil & gas wells Biological (living things) Respread topsoil (seed bank + microbes) Revegetate with native species N-fixers & mycorrhizal fungi rebuild soil Bioremediation — microbes clean contaminants Phytoremediation — plants draw up

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

They fix up the mine by cleaning it and planting some trees so it looks nice again and animals can come back to live there like before . "Cleaning it" isn't a named mechanical process — no reshaping, backfill or capping. "Planting trees" is on the right track but vague, and never linked to the goal. "Like before" wrongly implies full restoration; ignores safe/stable/sustainable.

✓ STRONG / MODEL ANSWER

A mechanical process is battering down (regrading) the steep highwalls and backfilling the void with overburden , which creates a stable landform that won't slump or collapse — making it safe and stable . A biological process is respreading stockpiled topsoil and revegetating with native species , whose roots bind the surface against erosion and rebuild a self-sustaining ecosystem — making the landform sustainable . Names a specific mechanical process AND its function (stability/safety). Names a specific biological process AND its function (sustainability). Explicitly ties both to the marked phrase "safe, stable and sustainable".

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.